

MINI AI ASSIGNMENT

Customer Review Sentiment Analysis

1. Assignment Overview

In this assignment, you will build a simple Artificial Intelligence / Machine Learning system that analyzes customer reviews and predicts whether each review is **Positive, Negative, or Neutral**. The goal is to understand how Natural Language Processing (NLP) can be used to analyze text.

2. Problem Statement

Companies receive thousands of customer reviews. Reading every review manually is time-consuming. Your task is to create a small sentiment analysis solution that automatically identifies the sentiment of customer reviews.

3. What You Need to Do

1. Collect or use a small dataset containing customer reviews.
2. Clean the review text by removing unnecessary symbols, spaces, or other unwanted text.
3. Assign or use sentiment labels such as Positive, Negative, and Neutral.
4. Convert text into numerical features using a simple NLP technique such as TF-IDF.
5. Train a basic machine learning model such as Logistic Regression or Naive Bayes.
6. Test the model using sample reviews.
7. Display the predicted sentiment for new customer reviews.
8. Write a short conclusion about the model's performance.

4. Suggested Dataset

For this mini assignment, you do not need a large dataset. You can create a CSV file yourself with around 20–30 reviews. Use two columns: **review** and **sentiment**.

review	sentiment
The product quality is excellent.	Positive
I loved the service and delivery was fast.	Positive
The product stopped working after two days.	Negative
Very poor customer support.	Negative
The package arrived yesterday.	Neutral
The product is okay for the price.	Neutral

5. Technology You Can Use

Purpose	Suggested Tool
Programming	Python

Data handling	Pandas
Text processing	NLP / basic preprocessing
Feature extraction	TF-IDF
ML model	Logistic Regression or Naive Bayes
Evaluation	Accuracy / Confusion Matrix
Environment	Google Colab or Jupyter Notebook

6. Expected Workflow

Customer Reviews → Text Cleaning → TF-IDF Features → ML Model → Sentiment Prediction → Evaluation

Students do not need to build a web application. A Google Colab/Jupyter Notebook with clear outputs is sufficient.

7. Minimum Notebook Requirements

1. Import the required Python libraries.
2. Load or create the review dataset.
3. Show at least 5 sample reviews.
4. Perform basic text preprocessing.
5. Split the data into training and testing sets.
6. Apply TF-IDF vectorization.
7. Train one classification model.
8. Predict sentiment for the test data.
9. Calculate accuracy.
10. Test the model on at least 3 new reviews written by you.

8. Example Final Output

Customer Review	Predicted Sentiment
Amazing product and fast delivery!	Positive
The quality is terrible.	Negative
The package arrived today.	Neutral

9. Questions Students Must Answer

1. What is sentiment analysis?
2. Why is sentiment analysis useful for companies?
3. What is NLP and why is it needed for this task?
4. What is TF-IDF?
5. Which machine learning model did you use and why?
6. What accuracy did your model achieve?
7. Give one example where the model may produce an incorrect prediction.

10. Submission Requirements

- Google Colab / Jupyter Notebook (.ipynb)
- Dataset (.csv), if you created a separate file
- Short conclusion (5–8 lines)
- Screenshots of the final predictions

11. Time Limit

Recommended completion time: 20–30 minutes. Students may use a small dataset and a simple model. The focus is on understanding the AI/NLP workflow, not on building a production-level application.

12. Evaluation – 25 Marks

Component	Marks
Dataset and preprocessing	5
TF-IDF / feature extraction	5
ML model and prediction	5
Evaluation and results	5
Explanation and conclusion	5
Total	25

13. Optional Bonus

For extra credit, students can compare two models (for example, Logistic Regression vs. Naive Bayes) and state which model performs better and why.

Learning Outcome: By completing this assignment, students will understand the basic workflow of NLP-based sentiment classification, including text preprocessing, feature extraction, machine learning, prediction, and evaluation.